Homework 3

MATH 441: DIFFERENTIAL EQUATIONS

NAME: DATE:

(Exercises are taken from *Elementary Differential Equations and Boundary Value Problems* by William E. Boyce and Richard C. DiPrima)

In each of Problems 1 and 2, transform the given initial value problem into an equivalent problem with the initial point at the origin

1. Exercise §2.8 #1.

$$dy/dt = t^2 + y^2, \qquad y(1) = 2$$

2. Exercise §2.8 #2.

$$dy/dt = 1 - y^3, \qquad y(-1) = 3$$

In each of Problems 4 and 5, let $\phi_0(t) = 0$ and define $\{\phi_n(t)\}$ by the method of successive approximations
(a) Determine $\{\phi_n(t)\}$ for an arbitrary value of n.
(b) Plot $\{\phi_n(t)\}$ for $n = 1, ..., 4$. Observe whether the iterates appear to be converging.
(c) Express $\lim_{n\to\infty} \phi_n(t) = \phi(t)$ in terms of elementary functons; that is, solve the given intial value problem.
(d) Plot $|\phi(t) - \phi_n(t)|$ for $n = 1, ..., 4$. For each of $\phi_1(t), ..., \phi_4(t)$, estimate the interval in which it is a reasonably good approximation to the actual solution.

3. Exercise §2.8 #4.

$$y' = -y - 1, \qquad y(0) = 0$$

4. Exercise §2.8 #5.

$$y' = -y/2 + t, \qquad y(0) = 0$$